

Andy Glover

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Engineering Portfolio: <https://aglover5.github.io>

EDUCATION

University of Southern California – Los Angeles, CA

M.S. Astronautical Engineering – 2025-2025

B.S. Aerospace Engineering – 2022–2024

Saint Mary's College of California – Moraga, CA

B.A. Liberal Arts (Engineering Dual-Degree) – 2018–2021

TECHNICAL SKILLS

Programming: MATLAB, Python (NumPy), VBA (Excel)

Controls & Optimization: CasADi, IPOPT

Simulation & Modeling: Siemens NX, MuJoCo

Hardware & Embedded Systems: Arduino, STM32, Garmin LIDAR-Lite v3

Experimental Methods: LabVIEW, Dye flow visualization

EXPERIENCE

Model Predictive Control for On-Orbit Mass Property Identification *Graduate Research Project* | Oct 2025 – Present

- Contributed to the development of an MPC framework (CasADi/IPOPT) for satellite agents performing excitation-based mass property identification of an unknown on-orbit structure.
- Led cost function design by implementing a tunable Fisher Information term, reducing mass estimation uncertainty from ~3–16% (uniform weighting) to ~1% through nonuniform, mass-prioritized tuning.
- Implemented and evaluated a time-decaying Fisher weighting heuristic to dynamically balance early-stage excitation and late-stage trajectory tracking.
- Currently extending the framework with sensor noise modeling and EKF-based estimation to improve robustness under a realistic on-orbit measurement environment.

Space Engineering Research Center (SERC) Volunteer *USC Information Sciences Institute* | Feb 2025 – May 2025

- Assisted the project lead for LEAPFROG lunar lander project as a part time contributor.
- Validated LidarLite V3 range reading functionality with Arduino, helped construct a manual kill switch circuit for JetCat P300-Pro RC Jet Engine with visual alert LED.

Senior Project: Aerodynamic Performance of Dragonfly-Inspired Airfoils *USC – AME 441: Senior Laboratory I July 2024 – Dec 2024*

- Designed two corrugated airfoils in Siemens NX, inspired by dragonfly wing CFD studies; tested against a NACA 2410 control in USC's water channel.
- Built an adjustable CAD mounting system using T-slot frame and vertical cantilever for strain-gauge-based lift measurement.
- Conducted dye flow visualization to capture vorticity in wing corrugations.
- Processed the strain gauge data to calculate lift coefficients and presented findings at the AME Senior Project Expo.

Aerial Drone CAD Design Project *USC – AME 308 I Nov 2023 – Dec 2023*

- Collaborated with my team to design a recreational drone in Siemens NX.
- Created a camera mount and battery pack components from measurements of real-life drone; created exploded assembly view and parts list with mass/material details.

NASA L'SPACE Mission Concept Academy: Venus Drone Design *NASA Lucy Student Pipeline Accelerator I Jan 2021 – May 2021*

- Selected for 12-week NASA engineering academy; contributed to the drone concept for Venus atmospheric survey.
- Calculated telecommunications frequency, solar panel requirements, and designed scientific payload packaging within a 196 lb constraint.
- Drafted PDR documentation and subsystem overviews; presented to NASA engineers and mentors.

Mars CubeSat Mission Design – Summer Research Program *Saint Mary's College of California I Project Manager I May 2020 – Aug 2020*

- Led a 5-member interdisciplinary team, including JPL geophysicists, to design a mock CubeSat mission for Martian dust profiling.
- Conducted telecommunications system design, selecting hardware for X-band to DSN compatibility.
- Modeled a 3W power link budget accounting for antenna gain, cable loss, and atmospheric attenuation.
- Proposed deployable reflectarray antennas for independent Deep Space Network (DSN) communication in a sun-synchronous orbit.
- Delivered final proposal to NASA/JPL scientists.

LEADERSHIP & COLLABORATION

- **Project Manager** – Mars CubeSat Mission Summer Research
- **Team Member** – NASA L'SPACE Academy (Team Oracle)